JINMING NIAN

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SUMMARY

Experienced research assistant and PhD candidate specializing in ML and AI, with a strong background in retrieval, NLP, retrieval augmented generation, and AI reasoning. Skilled in scientific programming with Python/PyTorch, with additional experience in software development and distributed systems. Actively publishing research on retrieval and LLM-related topics.

EDUCATION

Ph.D. in Computer Science and Engineering *September 2023 - Present*
Santa Clara University
Advisor: **Prof. Yi Fang**

M.S. in Computer Science and Engineering *September 2021 - June 2023*
Santa Clara University
Relevant Courses: Machine Learning, Deep Learning, Information Retrieval, Natural Language Processing, Reinforcement Learning

B.S. in Physics, Minor in Music *September 2017 - June 2021*
University of California, Davis

WORK EXPERIENCE

Research Assistant, Santa Clara University *April 2023 - Present*
- Evaluated the reranking effectiveness of LLMs based on their likelihood of generating correct answers for a given query and diverse document contexts, producing a synthetic dataset with quality comparable to human labels for training dense retrievers
- Developed methods using LLMs to generate controlled hallucinations from news articles, creating highly relevant questions that ask for information absent in the document, mimicking RAG scenarios where the retrieved document is unfit for the question
- Analyzed the usefulness of LLM internal attention matrices as features for training a document reranking model
- Developed web-based tools to facilitate human annotation, results aggregation, and inter-annotator agreement calculations

PAPERS & PRE-PRINTS

- **W-RAG: Weakly Supervised Dense Retrieval in RAG for Open-domain Question Answering** — arXiv
(Under review for *Journal of Web Semantics*)
Jinming Nian, Zhiyuan Peng, Qifan Wang, Yi Fang
- **ScopeQA: A Framework for Generating Out-of-Scope Questions for RAG** — arXiv (Under review)
Zhiyuan Peng, **Jinming Nian**, Alexandre Evfimievski, Yi Fang

PROJECTS

Atari Agent (ELEN 552, Reinforcement Learning) *November 2023 - December 2023*
- Trained an AI agent to play Pong using Reinforcement Learning methods including a Deep Q-learning Network, Agent memory for experience replay, fixed Q-target, and the epsilon-greedy strategy
- Achieved 89% win-rate against a programmed player with an average winning score of 21:15

Distributed Publish-Subscribe System (COEN 317, Distributed Systems) *August 2022 - December 2022*
- Built a distributed service in Java with high scalability and reliability, capable of hosting millions of clients
- Established milliseconds level asynchronous server-to-server communication via Java RMI
- Implemented efficient broadcasting mechanism using gossiping protocols, failure detector, leader election, concurrency control, replication and consistency protocols, and client side Publish-Subscribe functionalities

SKILLS

Programming Languages: Python, Java, C++, Swift, Go
Tools: PyTorch, Transformers, BEIR, ColBERT, Pyserini, SentenceTransformer, Git
Systems: AWS EC2 & RDS, GCP CloudSQL, MySQL, Oracle, Linux